

Steering Blind Algorithms: Exploring the Impact of Generative AI on the Role of Designers

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Abstract

Recent advancements in AI, particularly those focused on replicating creativity, have sparked ongoing discussions about their impact on the role of designers. This paper presents the findings from qualitative interviews with 17 design professionals focused on strategic design work, exploring their perceptions and experiences with AI tools, as well as their views on how their roles may evolve in the future. The results highlight the diverse ways in which designers are leveraging AI in their work as well as the resulting impact on their roles, emphasizing the continued significance of expert human designers in steering development toward meaningful outcomes. The findings underscore the continued importance of contextual awareness, interdisciplinary collaboration, and human-centered design in crafting meaningful solutions, and highlight the role of relational expertise in design that manifests through interactions.

CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); Empirical studies in HCI.

Keywords

design professionals, generative AI, designer-AI collaboration

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1 Introduction

Recent advancements in artificial intelligence (AI) are poised to reshape industries, with estimates suggesting that up to 60% of jobs could be impacted by AI [1]. Notably, developments in generative AI (GenAI) tools that can create high-quality content are challenging the notion that creativity is uniquely human, a shift that may be particularly unsettling for design professionals focused on creating novel and innovative solutions. While AI optimists

highlight the potential of the technology, its rapid, technology-driven development has sparked concerns about its societal impact. In response, researchers are increasingly advocating for human-centered AI (HCAI), emphasizing the need to design AI systems that enhance human capabilities rather than replace them, underlining the need for contextual understanding to take into account human concerns and skills [2]. In the field of design, this calls for a deeper exploration of how AI may impact creative industries, design roles, and specific tasks and processes [3–5]. Building on previous research, this study offers a holistic exploration of the evolving roles of designers focused on strategic design work, examining how the influence of AI tools and the changing demands of the industry are shaping their ability to contribute meaningfully. It explores how emerging AI technologies are transforming design work, focusing on designers' collaboration with AI in the creative process and the resulting impact on their roles. The research seeks to uncover opportunities for further developing AI, while highlighting the unique value designers continue to offer, emphasizing the irreplaceable aspects of human creativity and expertise.

2 Background

Design scholars often emphasize the intuitive aspects of a designer's sensemaking and approach to problem-solving that enable creative, innovative outcomes. Schön [6] for example, critiques the dominance of 'technical rationality' and the prioritization of the application of scientific theories to problem-solving, for their inadequacy in addressing complex and open-ended problems. He argues that a technically-driven approach views professional practice as a process of problem *solving*, which overlooks the essential process of problem *setting*. This initial stage involves defining the goals to be achieved and selecting the appropriate methods to reach them. With conflicting and unclear objectives, naming and framing the problem to be solved becomes a precondition for exercising technical expertise. Moreover, scholars emphasize the importance of an abductive, iterative approach in addressing open-ended problems. Cross [7] describes design as a search, where not all relevant information is predictable and must be discovered along the way. He argues that effective solutions to ill-defined design problems emerge as designers engage in sketching and learning, filtering out irrelevant information throughout the process. Indeed, goal communication is perceived as a key challenge in fostering effective collaboration between designers and AI tools, as AI relies on explicit goal definitions, which contrasts with the designer's iterative approach [8]. Designers often start without clearly defined goals, refining and updating them through trial and error as they gain new insights throughout the process [8]. Current tools reflect

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the tendency to undermine the value of “the wandering stage” of playing around with pre-mature ideas and how it contributes to the creative outcome [9].

Despite the reservations, AI may help address the limitations of traditional, human-intensive design, allowing designers to shift their focus to problem-finding [10]. Studies indicate potential for AI to support designers by automating repetitive tasks throughout the design process, freeing up time to focus on the creative process [3, 11, 12]. Given the different strengths of humans and machines, they may be combined to complement each other [13-15]. AI excels at processing vast amounts of information and automating routine tasks, that can assist, for example, with the laborious process of analyzing large amounts of qualitative data [16], a central method in design research. Humans, then, contribute essential skills, such as intuition and creativity, allowing them to navigate uncertainty, address new challenges, set objectives, select relevant data, and evaluate final solutions effectively [8, 13]. This synergy is particularly evident in creative endeavors and situations where human interpretation of meaning complements AI techniques [2].

The ongoing debate about the role these technologies should play in supporting human work is characterized by discussion about AI systems’ impact on human autonomy and agency [17, 18], an individual’s ability to exercise control over their life. As fundamental aspects of well-being, they should be carefully considered in the development of technology to avoid demotivation [19]. This underlines the importance of finding a balance between what kind of work is rewarding and what creates value in the short-term. While some view AI tools as assistants that primarily help humans with various tasks, others perceive AI as a partner with creative agency that can be valuable in offering creative input and suggestions that influence the designer’s decision-making process. Despite the concerns, co-creative systems show significant potential in design work, and AI-generated, creative output may be well suited for divergent phases of the design process, where AI can provide inspiration and insight by offering exposure to a wide range of resources and producing a variety of alternatives [3, 9, 11, 12, 20].

At the same time, the value of design expertise is increasingly recognized as AI becomes more deeply integrated into systems and applications. Human-centered design (HCD), for example, is suggested as a key approach to ensuring the responsible development of AI-enabled systems [21-23]. Furthermore, involving end-users through participatory design in AI projects may not only be valuable from a moral standpoint, but also from a pragmatic perspective, as end-users are experts in their lives and can share contextual insights [2], enabling the development of useful solutions that match user needs and gain acceptance [24, 25]. Indeed, research suggests that when designers are involved in the early problem discovery phase in AI projects, they apply design thinking to expand the scope of the problem, often leading to the identification of entirely new opportunities [3, 26]. Designers play a key role in facilitating alignment between diverse disciplines and stakeholders, using multidisciplinary skills to integrate varied expertise. This enables them to reframe problems, identify issues worth solving, and drive innovation in AI projects [5, 26], and highlights the need to expand AI research beyond the realm of UX design to include service and systems-level designers who are engaged in early, strategic design work [3, 26]. As AI continues to evolve, transforming design tools

and reshaping industry demands, a critical question arises: what unique value do designers still offer in this shifting landscape, and how can AI enhance their capabilities?

3 Methods

This qualitative study used semi-structured interviews to examine the experiences and perspectives of design professionals on the impact of AI on their work and future roles. The data was collected in the spring of 2024 from companies across Finland. Participants were purposively selected to ensure a diverse representation of design professionals with experience or interest in technology and AI, aiming to capture in-depth perspectives from those who had actively engaged with AI. While many participants had a background in technology, this was not a requirement. The sample focused on professionals with a strategic role in design, including senior-level designers and heads of design teams. Their responsibilities spanned concept design, user experience design, user research, consumer insight, service design, business strategy, innovation, and AI implementation. A total of 17 professionals from various sectors participated. The data was transcribed, anonymized, and thematically analyzed following the approach outlined by Braun and Clarke [27]. The data was coded using ATLAS.ti software, and the resulting codes were organized into themes and sub-themes, which are presented in the following section.

4 FINDINGS

4.1 AI Tools Can Support Both the Creative and Analytical Aspects of Design

Most participants felt confident that AI would not replace them soon, given the strategic nature of their roles. In addition to manual and technical design tasks, their work involves engaging with diverse stakeholders—management, colleagues, clients, and users—and focusing on human-centric tasks such as identifying meaningful problems, defining shared goals, and aligning teams. Designing concepts, leading projects, and making context-driven decisions require a deep understanding of the strategic priorities and problem constraints, which in turn helps evaluate the relevance of different data sources. Participants underscored that in most cases, there is no objectively *right* or *wrong* direction; decision-making is influenced by a range of factors that cannot be easily automated. Many had already adopted different AI tools and most participants expressed a strong willingness towards AI-driven automation of routine tasks to free up time for more complex tasks. Many also recognized AI’s potential to assist across various stages of the design process and reported that AI improved efficiency, enabling them to focus on creative and critical thinking, and human interaction. This more strategic work was also perceived as more enjoyable. Participants agreed that the true value of designers lies not in the artifacts they create, but in their ability to generate concepts and ideas. As P16 notes, streamlining the design process enables faster iteration and testing, accelerating the path to better outcomes:

“I don’t see any sense in a designer’s job to spend time on drawing boxes and moving them around a pixel at a time. It’s not the mechanical work that creates value, but the ideas and the end result.”

Table 1: Participant demographics. *Combined number of years spent studying and working. **ID: Industrial Design; UI: User Interface Design; UX: User Experience Design; SD: Service Design; BD: Business Design.

Label	Organization	Experience*	Discipline**
P1	Consultancy: business strategy	20+ years in design with 5+ years focus on AI projects	UX, SD
P2	Self-employed consultant	10 years in design with 5+ years focus on AI projects	ID, SD
P3	Consultancy: business design	3 years in design and 5+ years in computer science	UX, SD
P4	Consultancy: IT	10+ years in design	UX, SD
P5	Company: technology	10+ years in design and 15+ years in business strategy	SD, BD
P6	Corporation: IT	10+ years in design	UX, SD
P7	Consultancy: design	25+ years in design	UX, SD
P8	Consultancy: business design	15+ years in design	ID, SD
P9	Company: technology	20+ years in design	ID, UX
P10	Consultancy: business	4 years in design and 10+ years in business	SD, BD
P11	Corporation: technology	10+ years in design with 2 years focus on AI projects	UX, SD
P12	Corporation: technology	10+ years in design with 4 years focus on AI projects	ID, UX
P13	Consultancy: IT	20+ years in design	UX, SD
P14	Consultancy: business	20+ years in design	UX, BD
P15	Corporation: industrial manufacturing	10+ years in design	ID, UX
P16	Startup: technology	20+ years in design with 5+ years focus on AI projects	UI, UX
P17	Corporation: ICT	5+ years in design and 5+ years in computer science	UX, SD

In addition to efficiency, participants reported that AI tools enhanced the quality of their work by providing access to valuable information, offering a more structured approach to work, and encouraging thinking outside the box. AI tools were valued for their ability to inspire, jumpstarting the ideation process and providing a foundation to build upon. The tools were found useful in the early stages of projects, supporting research, drafting, and initiating ideation, as well as for editing and finalizing their own creations across various contexts and purposes. Some highlighted the potential of AI tools to support design-driven decision-making by scaling research efforts and providing quantitative evidence to back qualitative insights, as P8 explains:

“AI has allowed us to combine a quantitative approach to our qualitative approach based on deep human understanding, which at the end of the day is our core know-how. Combining different research methods has worked really well and we’ve been able to provide proof for our findings from qualitative research more convincingly.”

Depending on the context, AI tools were described either as creative collaborators as well as *just* tools. The data showed the following three types of interaction with AI, each with a different perception of AI’s role based on its intended use:

AI for creative support: When AI was used for ideation, storytelling, and drafting, it was viewed as a creative partner that introduced fresh ideas the designer may not have thought of. AI supported the early stages of design by offering inspiration and diverse perspectives, helping to kick-start the creative process without preconceived expectations for the final output. In these divergent phases, designers did not expect final results but valued AI for generating ideas to build upon, with hallucinations even seen as an asset.

AI as a tool: When AI was used for research activities or refining their own work, participants emphasized that it is just a tool, with the designer responsible for ensuring output quality and validity. This applied when preparing materials with established best practices, editing content, or creating summaries, where the designer had clear expectations for the results. They valued AI for its efficiency and attention to detail. In this context, skilled use and effective prompting were key to achieving quality output, with the designer’s expertise and ability to specify prompts and evaluate output playing a crucial role. Lack of control and transparency were perceived as issues, and users remained cautious of potential errors or biases, for example, if they were using AI to support with initial desktop research or the preparation of research materials.

AI as a collaborator: This scenario combines the previous two, with AI used in a highly interactive, collaborative way. The designer uses AI to exchange ideas, refine concepts, and receive feedback, resembling working with a colleague. The designer provides direction, while AI offers creative input and critical analysis to help develop and improve ideas.

Despite the perceived benefits of AI, participants stressed that their own expertise and critical reflection were essential in leveraging the tools in a creative and responsible way. They emphasized that creative outcomes evolve through a process of trial and error, adaptation, and refinement of results, underlining the continued need for humans to lead the design process. P16 describes the iterative process:

“Getting to the final result still requires the process. If we had a text box or voice command for prompting in Photoshop or any other creative tool, you would still need to describe what you want to achieve. And I don’t think anyone can describe a creative outcome in one sentence. At least I can’t. It’s always about starting somewhere, and the final result is always an outcome of iteration.”

Participants agreed that familiarizing themselves with AI tools and recognizing both the strengths and limitations is essential for effectively leveraging AI as a tool as well as a design material. Nevertheless, lack of transparency and best practices to leverage the technology often left them confused and hoping for support.

4.2 Design Principles that Continue to Create Value in the Prevalence of AI

Participants with experience in developing AI-enabled solutions viewed the following core principles of design as key strengths that would remain relevant in the future.

4.2.1 A Problem-Oriented Approach Supports Overcoming AI Hype. Many participants shared experiences where the hype around AI and a technology-driven approach had led to AI being pursued simply for the sake of adopting the new technology without a clear understanding of how it can address specific challenges. They stressed the importance of taking a problem-oriented approach to avoid building potentially ineffective solutions. A designer’s mindset and methods were considered valuable in gaining a deeper understanding of underlying problems. While striving for shared understanding and effective collaboration was emphasized, some argued that to develop novel and innovative approaches, designers must often challenge the status quo, pushing for change and new perspectives. P4 explains:

“A key principle of mine is to go somewhat against the grain and take the time to precisely understand what we are truly trying to achieve and what the underlying drivers are in this process. We need to know what they are and take them into account to ensure we can create the best possible solution, even if it’s different from what was initially requested.”

4.2.2 HCD Supports the Design of Ethical and Useful Solutions. Human-centricity was seen as key to developing ethical and useful solutions. While participants stressed that responsible AI systems should be a shared concern, they believed designers are well-positioned to champion these concerns. Participatory methods proved valuable in uncovering user needs and values and developing solutions that gain users’ acceptance and are adopted by them. They agreed that while technical tasks can be automated, designers will remain essential in advocating for ethical design and addressing the needs of diverse user groups. P9 shares:

“It’s the designer’s task to be the spokesperson for human beings and the environment, because the economist will not do the job and the engineer may not have sufficient capabilities or interest for it. So, it’s the designer’s responsibility.”

4.2.3 Interdisciplinary Skills Are Increasingly Valuable for Enabling Shared Visions. Participants saw their role in AI projects as that of a translator that demystifies AI, clarifying its potential and feasibility for diverse stakeholders. Their interdisciplinary backgrounds enabled them to effectively translate technology into more approachable terms for stakeholders that lack AI knowledge, and to communicate client and user needs back to engineers and data scientists, enabling the development of meaningful solutions. Overall, strong communication and interpersonal skills were deemed essential for aligning teams and navigating varying motivations. Participants elaborate:

“You have different experts in a team, so as a designer, one of my biggest tasks is to put everyone on the same page regarding what we are doing. And then we define the problem in a proper way, while at the same time questioning things on a high level at the very beginning. We sort of guide and facilitate the direction of the development in projects.” [P4]

“As a designer you have an increased role of a translator in projects involving AI compared to traditional software projects, where the vision is often clearer. Our role is to create a vision and act as a translator between technical specialists, our client, the end users, and other stakeholders. That kind of work has become more central to my role, that we clarify the vision and speak technical language with the technical experts and the client’s own language with them. I think the ability to speak in different levels of precision in different contexts is part of a designer’s tool kit.” [P12]

4.2.4 Contextual Knowledge Allows Driving Projects Towards Meaningful Directions. AI’s limitations highlight the essential role of designers in understanding the broader context of their work and guiding projects toward desired outcomes. While AI can enhance specific tasks, participants emphasized that they are responsible for ensuring results meet expectations and create value. This involves maintaining a holistic perspective, identifying key problems, balancing constraints, and aligning teams and stakeholders with potentially conflicting values and objectives. Participants stressed that AI cannot navigate the complexity of working with clients and stakeholders, and that AI’s lack of intention and care make humans

responsible for driving development towards meaningful outcomes. Participants reflect:

“As a designer, your main task is to guide and have an overview of the big picture, bringing the different perspectives to the table and shaping AI generated output, while at the same time learning from the different perspectives.” [P12]

“I think humans will be needed to drive the design process forward because they are the ones who care about making it happen. AI on itself doesn’t want to do anything and the designer’s task is to define what needs to be done. The designer will always be needed to drive projects, direct where the project is headed, and define what needs to be done next.” [P3]

5 DISCUSSION AND FUTURE WORK

The data suggests that automation and augmentation are interdependent in design work [14], and mindfully implemented, AI can augment human agency rather than diminish it [28] by freeing designers to work on more complex tasks. The data aligns with existing research, highlighting how AI can enhance both the efficiency and quality of designers’ output. Furthermore, the participants’ willingness to adopt AI reflects their recognition of the limitations they face in solving complex problems. Especially AI’s ability to support divergent thinking and offer alternative perspectives aligns with designers’ curiosity to deeply understand problems and develop holistic solutions [29, 30]. A design-driven approach has been shown to mitigate cognitive bias and overconfidence by supporting exposure to diverse perspectives, fostering a more holistic and objective approach [31], and AI’s potential to streamline research, access vast data, and support more objective insights may further enhance this effort.

As AI takes over and supports technical design tasks, including prototyping and visualization, the designer’s mentality [32] becomes a cornerstone of design expertise. Artifacts, while still important for communication and facilitating the design process, primarily serve to create meaning [33]. This suggests that a designer’s agency doesn’t rely on personally creating every artifact, such as workshop materials or presentation slides, but on understanding the underlying logic to oversee the process, making transparency of AI tools essential. However, in situations where an artifact is central to the final design solution, such as a brand or product concept, designers may prefer more control to ensure originality and attention to detail. The different AI use scenarios support the claim that AI’s role in collaborative settings should be customizable, tailored to different situations and individual needs [9]. The study calls for further enquiry of designer-AI interactions across creative and research tasks to optimize AI support in the different phases of the design process.

While AI-driven augmentation may indeed enable designers to focus on problem-finding [10], the data aligns with existing literature, highlighting that problem-finding, framing, and goal-setting are iterative processes. Designers continually refine problems and adjust objectives as they uncover new information, enhancing their ability to address core issues [8, 9], emphasizing the need to keep designers in the loop. This is particularly relevant when solving

wicked problems with conflicting objectives that require understanding diverse stakeholders’ needs and values, and developing hypotheses that can be explored further [34].

The findings indicate that expertise in design should not be seen merely as an intellectual possession, but as something shaped through interactions [35]. This *relational expertise* emerges through collaboration with other actors or clients, and is applied to specific cases or problems, often requiring an understanding of the cultural and political context [35]. Participants’ willingness to adopt AI tools reflects the high level of relational expertise required in strategic design, which involves collaborating with clients and evaluating diverse sources of information [3, 35]. AI-driven automation can free designers to focus their efforts on facilitating interdisciplinary collaboration and stakeholder engagement, which are crucial for identifying opportunities [3, 26] and developing AI systems in a human-centered way [21, 23]. Since AI systems both lack intentionality and are not neutral, but reflect the world view of their creators as well as reproduce societal prejudices and inequalities [2, 36], designers highlight their responsibility in steering development toward responsible, meaningful outcomes. This responsibility applies both to the use of AI tools and the design of AI-enabled solutions. They bring invaluable contextual awareness and tacit knowledge gained through experience [13, 14, 37, 38], allowing them to define meaningful goals and balance diverse stakeholder constraints.

Overall, as design becomes more strategic, involving vision alignment and team negotiation, strong communication and interpersonal skills are considered valuable. Designers may leverage empathy and intuition to excel in these “feeling tasks” that are becoming increasingly important [39] to complement AI tools. While the data underscores the growing need for effective collaboration, it suggests that a designer’s role remains critical in questioning assumptions, ensuring transparency, and prioritizing human-centered values, such as user experience and accessibility, which may be considered disruptive in a capitalistic, tech-driven environment. When recognized and supported, the designer’s role may develop from form-giving to shaping strategic priorities and setting long-term goals, enabling organizations to progress along the Danish Design Ladder [40] and leverage design more strategically. This highlights the need for exploration of the impact of early designer involvement on AI project outcomes and means of supporting it.

While AI systems are rapidly developing and may soon overcome some of their current shortcomings, including limited contextual awareness and ethical reasoning, the findings nevertheless highlight the need to question AI’s ability to optimize for the *best* outcome. The data reflects the notion that especially complex problems involving conflicting values and expectations can never be completely solved, but need to be continuously resolved as any solution will generate waves of consequences that may lead to undesirable outcomes [41]. Therefore, solutions will always be partial and subject to ongoing refinement. This implies that regardless of how much AI may take over tasks, since there are no objectively right solutions there will always be a need for professional designers to facilitate the design process and provide guidance. Designers may continue to play a crucial part in navigating conflicting values and needs of different stakeholders, bringing in diverse perspectives, negotiating strategic priorities, aligning people, and leading change.

6 CONCLUSION AND LIMITATIONS

This study explores how the growing prevalence of AI is reshaping the role of designers. It elaborates how designers use AI to enhance work efficiency and quality, enabling them to focus on applying their expertise in HCD, interdisciplinary skills, and contextual awareness, to identify meaningful problems and develop valuable solutions. The study builds on existing research by focusing on strategic level designers, looking at their role development holistically. The study reveals how the effective use of AI tools allows designers to shift their effort onto more complex tasks and highlights the continued value of expertise in design. The findings underline the importance of relational expertise that enables professional designers to navigate complexity in their work. The study's limitations include its broad scope aimed at providing an extensive overview and avenues for further research, that however, lacks depth in the specific areas. Additionally, by focusing on senior-level professionals in strategic roles, the insights may not reflect the perspectives of designers at other career stages. Lastly, the fast-evolving nature of AI technology, participants' varying understanding of its potential, and concerns about protecting their profession should also be considered. Nevertheless, the potential influence of participants' personal opinions about AI on the data should not be viewed as a limitation alone, as an understanding of design professionals' values and experiences is crucial for guiding the ethical and responsible development of AI [2].

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